TEST HANDOFF | WINDOWS XP SP3 X86

Ruisi USB DLP

Installation and Hardware Qualification Manual

Package	Target	Status
1.1.0	Windows XP Professional SP3, x86, NTFS	UNVERIFIED compatibility build

> **LAB USE ONLY:** This package enables live default-deny USB enforcement. Use only in an isolated test environment, on disposable test media, and from a VM snapshot. It is unsigned and is not approved for production deployment.

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1. Purpose and supported scope

This manual gives an external qualification team a repeatable procedure for installing the native XP server and workstation agent, proving default-deny behavior with physical USB storage, exercising request approval and revocation, and returning usable evidence to the development team.

- Supported target: 32-bit Windows XP Professional SP3 installed on NTFS.
- Unsupported targets: XP x64, XP before SP3, and any newer Windows release using this XP-specific MSI.
- Server and agent are separate products. A small lab may install both on one XP VM; a representative test uses separate server and workstation VMs.
- The agent installs with DryRun=0 and default deny. Unapproved USB storage can be disabled immediately after installation.
- The legacy transport is HTTP with Basic authentication. It is permitted only on an isolated test network.

> **Qualification boundary:** The build, MSI installation, service startup, enrollment, and policy download have been exercised on XP. Physical USB blocking, notifier behavior, approval, revocation, service-down behavior, and multi-volume media remain external hardware qualification items.

2. Package contents

Path	Purpose
client\RuisiUsbDlpXpAgent-1.1.0-x86.msi	Native XP agent and on-demand notifier
client\install-client.cmd	Validated live-mode agent installation wrapper
server\RuisiUsbDlpXpServer-1.1.0-x86.msi	Native XP management/API server
server\install-server.cmd	Validated server configuration and installation wrapper
server\admin-server.cmd	Safe local request/approval/revocation wrapper
config\deployment.ini	Editable server URL, port, and profile
config\enrollment-token.txt	Shared bootstrap token; replace before installation
config\admin-password.txt	Administrator password; replace before installation
SHA256SUMS	Integrity hashes for every file in the kit

3. Lab preparation

1. **Create restore points.** Take powered-off VM snapshots of the XP server and workstation before installation.
2. **Isolate the network.** Use a host-only or otherwise isolated test segment. Do not bridge port 5080 to an untrusted network.
3. **Prepare test media.** Use a USB stick with no valuable data. Safely eject it and leave it disconnected during installation.
4. **Confirm prerequisites.** Verify XP Professional SP3, 32-bit architecture, NTFS system volume, local Administrator access, and correct date/time.
5. **Prepare passthrough.** Verify the hypervisor can attach the physical USB device to XP before relying on any DLP result.
6. **Copy and verify.** Copy the entire kit to each VM. Verify SHA256SUMS on a modern trusted machine before execution.

> **USB passthrough:** A hypervisor capture failure is not an agent result. First prove that XP receives a USB or USBSTOR Plug and Play node. VirtualBox on some Windows 11 xHCI hosts may claim a device without delivering it to the guest; a USB 2.0 hub or different host may be required.

4. Configure the kit

4.1 Replace both placeholders

- Replace config\enrollment-token.txt with one random ASCII line of at least 32 characters. The server and every workstation in this test deployment must use the identical file.
- Replace config\admin-password.txt with one unique ASCII line of at least 14 characters. Do not reuse a production password.
- Keep both files access-controlled. Do not paste their values into command lines, screenshots, tickets, or test logs.

> **Important:** The shipped values are deliberate placeholders and must not be used as credentials.

4.2 Edit deployment.ini

Key	Required value
ServerUrl	HTTP URL reachable from the XP workstation, for example http://192.168.77.20:5080
Port	Server listener port; keep 5080 unless the isolated lab requires another port
AdminUsername	admin
Profile	legacy-xp-x86; do not change

```
[Deployment]
ServerUrl=http://192.168.77.20:5080
Port=5080
AdminUsername=admin
Profile=legacy-xp-x86
```

5. Install the server

1. **Open an elevated prompt.** Sign in as a local Administrator and open Command Prompt.
2. **Open the server folder.** Change directory to the kit's server folder.
3. **Run the wrapper.** Execute install-server.cmd. Do not launch the MSI directly because the wrapper validates and protects the configuration files.
4. **Open the firewall only in the lab.** If client and server are separate machines, allow the configured TCP port for the isolated subnet.
5. **Verify the service.** Confirm UsbDlp.LegacyServer reports STATE 4 RUNNING.

```
cd C:\RuisiUsbDlp-legacy-xp-x86-1.1.0\server
install-server.cmd
netsh firewall add portopening protocol=TCP port=5080 name="Ruisi USB DLP Lab"
mode=ENABLE scope=SUBNET
sc query UsbDlp.LegacyServer
```

> **Expected result:** The wrapper prints that the server was installed on the configured HTTP port. The service is RUNNING and the data directory is protected under All Users Application Data\UsbDlp.

6. Install the workstation agent

1. **Keep USB storage disconnected.** No removable test media should be attached while live enforcement is installed.
2. **Confirm server reachability.** From the workstation, ping the server address and confirm the isolated firewall path is correct.
3. **Run the wrapper as Administrator.** Execute client\install-client.cmd. The wrapper installs the exact XP MSI, protects agent data, stages the token, writes DryRun=0, and restarts the service.

4. **Verify service and enrollment.** Confirm the agent is RUNNING, then inspect the log for dry_run=false, enrollment_succeeded, and policy_applied version=1.

```
cd C:\RuisiUsbDlp-legacy-xp-x86-1.1.0\client
install-client.cmd
sc query UsbDlp.LegacyAgent
type "C:\Documents and Settings\All Users\Application Data\UsbDlp\legacy-agent.log"
```

> **Fail closed at startup:** The agent begins with default deny even before a policy is downloaded. A missing server does not turn the agent into dry-run mode.

7. Execute the core hardware workflow

7.1 Block and request access

1. **Start evidence capture.** Record time, VM snapshot name, USB manufacturer/model, VID/PID, serial number if available, and whether the device has multiple partitions.
2. **Insert the unapproved USB stick.** Attach it to the XP workstation through the hypervisor. Do not open or write files.
3. **Verify the block.** Within the expected interval, the notifier should appear with device details and no usable removable drive should remain.
4. **Submit a request.** Select 1, 7, 30, or 90 days; enter a 10-2000 character business justification; select Submit request.
5. **Verify dismissal.** The notifier should report submission and close automatically. A duplicate pending request should state that a recent request exists and also close.

```
type "C:\Documents and Settings\All Users\Application Data\UsbDlp\legacy-agent.log"
rem Expected: blocked instance_id=USBSTOR\...
```

7.2 Approve the request

1. **List requests on the server.** Run server\admin-server.cmd --requests from an elevated prompt.
2. **Copy the pending request ID.** Match the workstation, device, and user; do not approve an unrelated request.
3. **Approve.** Run admin-server.cmd --approve REQUEST_ID 30, replacing REQUEST_ID with the exact value.
4. **Wait for convergence.** Allow up to one 30-second policy poll. If XP does not surface the device while attached, unplug and replug it once.
5. **Verify use.** The device should be enabled and its expected partition or partitions should become accessible.

```
admin-server.cmd --requests
admin-server.cmd --approve REQUEST_ID 30
```

> **Why the wrapper restarts the server:** The local compatibility CLI updates encrypted persisted state. admin-server.cmd briefly stops the service, applies the command, and restarts it so agents immediately receive the new policy version.

7.3 Revoke and deny

1. **Revoke the approved request.** Run admin-server.cmd --revoke REQUEST_ID.
2. **Wait for policy convergence.** Allow up to 30 seconds. If necessary, unplug and replug once.
3. **Verify renewed blocking.** No partition from the revoked device should remain usable. Multi-volume media must expose no usable partition.
4. **Exercise denial.** Create a fresh pending request with another device or after test-state reset, then run admin-server.cmd --deny REQUEST_ID and confirm access remains blocked.

```
admin-server.cmd --revoke REQUEST_ID
admin-server.cmd --deny REQUEST_ID
```

8. Acceptance checklist

[ ] Both MSI files install only on XP SP3 x86 and reject the wrong OS profile.

[ ] Server and agent services are RUNNING after fresh install and after rerunning each wrapper.

[ ] Agent log reports dry_run=false, default_deny=true, enrollment_succeeded, and policy_applied.

[ ] An unapproved USB storage device is disabled and exposes no usable partition.

[ ] The notifier appears only for a newly blocked device and displays friendly name, instance ID, VID, and PID.

[ ] Submit remains disabled for fewer than 10 justification characters and works at 10-2000 characters.

[ ] The notifier closes after successful or duplicate-pending submission.

[ ] Approval is scoped to the exact workstation and device instance and converges within one policy poll or one replug.

[ ] Revocation blocks the attached device or blocks it on the first replug after policy convergence.

[ ] Multi-volume USB media exposes no usable partition while blocked.

[ ] Keyboard, mouse, webcam, audio, internal disks, hubs, and controllers remain usable.

[ ] Stopping the XP agent is recorded as legacy_backstop_unavailable; no full service-down fail-closed claim is made.

[ ] No password or enrollment token appears in process listings, command output, or collected logs.

9. Diagnostics and evidence collection

Evidence	Location or command
Agent log	C:\Documents and Settings\All Users\Application Data\UsbDlp\legacy-agent.log
Agent configuration	...\Application Data\UsbDlp\legacy-agent.ini; redact paths if needed
Services	sc query UsbDlp.LegacyAgent and sc query UsbDlp.LegacyServer
Device state	Device Manager screenshots before insertion, blocked, approved, and revoked
Requests	admin-server.cmd --requests
Workstations	admin-server.cmd --workstations
Windows logs	Event Viewer System and Application entries at the test timestamps
Package integrity	SHA-256 verification output for SHA256SUMS

> **Secret handling:** Do not collect admin-password.txt, enrollment-token.txt, bootstrap-token.txt, identity.dat, policy.dat, or server-state.dat in an ordinary ticket. If encrypted state is required, transfer it through an approved secure channel.

- Record exact timestamps and time zone for each insertion, request, decision, and observed state change.
- Record whether the physical device reached XP. A missing USB or USBSTOR node is a hypervisor passthrough failure, not a DLP enforcement result.
- For each failure, include the previous successful step and whether unplug/replug, service restart, or VM restart changed the outcome.
- Return the completed acceptance checklist, screenshots, redacted logs, package hash results, VM details, and device inventory.

10. Uninstall and rollback

1. **Disconnect test media.** Safely eject or unplug all removable storage before uninstalling.
2. **Uninstall the agent.** Use Control Panel > Add or Remove Programs > Ruisi USB DLP Legacy Agent.
3. **Uninstall the server.** Use Control Panel > Add or Remove Programs > Ruisi USB DLP Legacy Server.
4. **Verify services are gone.** Run both sc query commands and confirm each named service no longer exists.
5. **Preserve evidence.** The protected Application Data directory may remain for audit and reinstall. Do not delete it until evidence has been secured.

6. **Restore the snapshot.** For a clean test rerun, restore the powered-off pre-install VM snapshots.

> **XP service-down limitation:** XP lacks the modern layered installation backstop. A newly attached device can become usable while the agent service is stopped. This limitation is expected and must be reported, not hidden or described as fully fail-closed.

11. Test report summary

Field	Tester entry
Tester / organization	
Date, time zone	
XP edition / service pack	
Hypervisor / version	
Server and workstation topology	
USB device model / VID / PID / serial	
Single- or multi-volume	
Overall result	PASS / FAIL / BLOCKED BY LAB
Failed checklist item IDs	
Evidence archive name / SHA-256	

End of manual